



INTALG: Radicals and Exponents

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Learning Objectives

- Simplify square roots of perfect squares
- Apply the product and quotient rules for exponents
- Convert between radical and rational exponent form
- Simplify expressions with negative exponents

Simplify each expression completely. Show all steps and circle your final answer.

Exponent rules

1. Simplify: $x^7 \cdot x^2$.

$$x^7 \cdot x^2$$

Answer: _____

2. Simplify: x^4 / x^1 .

$$x^4 \div x^1$$

Answer: _____

3. Simplify: $x^2 \cdot x^5$.

$$x^2 \cdot x^5$$

Answer: _____

4. Simplify: x^4 / x^3 .

$$x^4 \div x^3$$

Answer: _____

5. Simplify: $x^5 \cdot x^4$.

$$x^5 \cdot x^4$$

Answer: _____

6. Simplify: x^7 / x^2 .

$$x^7 \div x^2$$

Answer: _____



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7. Simplify: $x^3 \cdot x^5$.

$$x^3 \cdot x^5$$

Answer: _____

8. Simplify: x^3 / x^1 .

$$x^3 \div x^1$$

Answer: _____

9. Simplify: $x^6 \cdot x^3$.

$$x^6 \cdot x^3$$

Answer: _____

10. Simplify: x^3 / x^2 .

$$x^3 \div x^2$$

Answer: _____

11. Simplify: $x^6 \cdot x^3$.

$$x^6 \cdot x^3$$

Answer: _____

12. Simplify: x^8 / x^4 .

$$x^8 \div x^4$$

Answer: _____

13. Simplify: $x^2 \cdot x^4$.

$$x^2 \cdot x^4$$

Answer: _____

14. Simplify: x^2 / x^1 .

$$x^2 \div x^1$$

Answer: _____

15. Simplify: $x^2 \cdot x^4$.

$$x^2 \cdot x^4$$

Answer: _____



16. Simplify: x^8 / x^3 .

$$x^8 \div x^3$$

Answer: _____

Square roots

17. Evaluate: $\sqrt{64}$.

$$\sqrt{64}$$

Answer: _____

18. The area of a square is 49 sq ft. What is the side length? Find $\sqrt{49}$.

$$\sqrt{49}$$

Answer: _____

19. Evaluate: $\sqrt{100}$.

$$\sqrt{100}$$

Answer: _____

20. The area of a square is 16 sq ft. What is the side length? Find $\sqrt{16}$.

$$\sqrt{16}$$

Answer: _____

21. Evaluate: $\sqrt{36}$.

$$\sqrt{36}$$

Answer: _____

22. The area of a square is 225 sq ft. What is the side length? Find $\sqrt{225}$.

$$\sqrt{225}$$

Answer: _____

23. Evaluate: $\sqrt{225}$.

$$\sqrt{225}$$

Answer: _____



24. The area of a square is 9 sq ft. What is the side length? Find $\sqrt{9}$.

$$\sqrt{9}$$

Answer: _____

25. Evaluate: $\sqrt{36}$.

$$\sqrt{36}$$

Answer: _____

26. The area of a square is 121 sq ft. What is the side length? Find $\sqrt{121}$.

$$\sqrt{121}$$

Answer: _____

27. Evaluate: $\sqrt{81}$.

$$\sqrt{81}$$

Answer: _____

28. The area of a square is 100 sq ft. What is the side length? Find $\sqrt{100}$.

$$\sqrt{100}$$

Answer: _____

29. Evaluate: $\sqrt{169}$.

$$\sqrt{169}$$

Answer: _____

30. The area of a square is 9 sq ft. What is the side length? Find $\sqrt{9}$.

$$\sqrt{9}$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Square roots, Exponent rules. All answers verified by independent computation.

Solutions



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Exponent rules

1. Simplify: $x^7 \cdot x^2$.

$$x^7 \cdot x^2$$

→ Product rule: $x^7 \cdot x^2 = x^{(7+2)} = x^9$.

Answer: x^9

2. Simplify: x^4 / x^1 .

$$x^4 \div x^1$$

→ Quotient rule: $x^4 / x^1 = x^{(4-1)} = x^3$.

Answer: x^3

3. Simplify: $x^2 \cdot x^5$.

$$x^2 \cdot x^5$$

→ Product rule: $x^2 \cdot x^5 = x^{(2+5)} = x^7$.

Answer: x^7

4. Simplify: x^4 / x^3 .

$$x^4 \div x^3$$

→ Quotient rule: $x^4 / x^3 = x^{(4-3)} = x^1$.

Answer: x^1

5. Simplify: $x^5 \cdot x^4$.

$$x^5 \cdot x^4$$

→ Product rule: $x^5 \cdot x^4 = x^{(5+4)} = x^9$.

Answer: x^9

6. Simplify: x^7 / x^2 .

$$x^7 \div x^2$$

→ Quotient rule: $x^7 / x^2 = x^{(7-2)} = x^5$.

Answer: x^5

7. Simplify: $x^3 \cdot x^5$.

$$x^3 \cdot x^5$$

→ Product rule: $x^3 \cdot x^5 = x^{(3+5)} = x^8$.

Answer: x^8



8. Simplify: x^3 / x^1 .

$$x^3 \div x^1$$

→ Quotient rule: $x^3 / x^1 = x^{(3-1)} = x^2$.

Answer: x^2

9. Simplify: $x^6 \cdot x^3$.

$$x^6 \cdot x^3$$

→ Product rule: $x^6 \cdot x^3 = x^{(6+3)} = x^9$.

Answer: x^9

10. Simplify: x^3 / x^2 .

$$x^3 \div x^2$$

→ Quotient rule: $x^3 / x^2 = x^{(3-2)} = x^1$.

Answer: x^1

11. Simplify: $x^6 \cdot x^3$.

$$x^6 \cdot x^3$$

→ Product rule: $x^6 \cdot x^3 = x^{(6+3)} = x^9$.

Answer: x^9

12. Simplify: x^8 / x^4 .

$$x^8 \div x^4$$

→ Quotient rule: $x^8 / x^4 = x^{(8-4)} = x^4$.

Answer: x^4

13. Simplify: $x^2 \cdot x^4$.

$$x^2 \cdot x^4$$

→ Product rule: $x^2 \cdot x^4 = x^{(2+4)} = x^6$.

Answer: x^6

14. Simplify: x^2 / x^1 .

$$x^2 \div x^1$$

→ Quotient rule: $x^2 / x^1 = x^{(2-1)} = x^1$.

Answer: x^1



15. Simplify: $x^2 \cdot x^4$.

$$x^2 \cdot x^4$$

→ Product rule: $x^2 \cdot x^4 = x^{(2+4)} = x^6$.

Answer: x^6

16. Simplify: x^8 / x^3 .

$$x^8 \div x^3$$

→ Quotient rule: $x^8 / x^3 = x^{(8-3)} = x^5$.

Answer: x^5



Square roots

17. Evaluate: $\text{sqrt}(64)$.

$$\sqrt{64}$$

→ Find the number that multiplied by itself equals 64.

→ $\text{sqrt}(64) = 8$.

Answer: $\sqrt{64} = 8$

18. The area of a square is 49 sq ft. What is the side length? Find $\text{sqrt}(49)$.

$$\sqrt{49}$$

→ $\text{Side} = \text{sqrt}(\text{Area}) = \text{sqrt}(49) = 7 \text{ ft.}$

Answer: $\sqrt{49} = 7$

19. Evaluate: $\text{sqrt}(100)$.

$$\sqrt{100}$$

→ Find the number that multiplied by itself equals 100.

→ $\text{sqrt}(100) = 10$.

Answer: $\sqrt{100} = 10$

20. The area of a square is 16 sq ft. What is the side length? Find $\text{sqrt}(16)$.

$$\sqrt{16}$$

→ $\text{Side} = \text{sqrt}(\text{Area}) = \text{sqrt}(16) = 4 \text{ ft.}$

Answer: $\sqrt{16} = 4$

21. Evaluate: $\text{sqrt}(36)$.

$$\sqrt{36}$$

→ Find the number that multiplied by itself equals 36.

→ $\text{sqrt}(36) = 6$.

Answer: $\sqrt{36} = 6$

22. The area of a square is 225 sq ft. What is the side length? Find $\text{sqrt}(225)$.

$$\sqrt{225}$$

→ $\text{Side} = \text{sqrt}(\text{Area}) = \text{sqrt}(225) = 15 \text{ ft.}$

Answer: $\sqrt{225} = 15$



23. Evaluate: $\sqrt{225}$.

$$\sqrt{225}$$

→ Find the number that multiplied by itself equals 225.

→ $\sqrt{225} = 15$.

Answer: $\sqrt{225} = 15$

24. The area of a square is 9 sq ft. What is the side length? Find $\sqrt{9}$.

$$\sqrt{9}$$

→ $\text{Side} = \sqrt{\text{Area}} = \sqrt{9} = 3 \text{ ft.}$

Answer: $\sqrt{9} = 3$

25. Evaluate: $\sqrt{36}$.

$$\sqrt{36}$$

→ Find the number that multiplied by itself equals 36.

→ $\sqrt{36} = 6$.

Answer: $\sqrt{36} = 6$

26. The area of a square is 121 sq ft. What is the side length? Find $\sqrt{121}$.

$$\sqrt{121}$$

→ $\text{Side} = \sqrt{\text{Area}} = \sqrt{121} = 11 \text{ ft.}$

Answer: $\sqrt{121} = 11$

27. Evaluate: $\sqrt{81}$.

$$\sqrt{81}$$

→ Find the number that multiplied by itself equals 81.

→ $\sqrt{81} = 9$.

Answer: $\sqrt{81} = 9$

28. The area of a square is 100 sq ft. What is the side length? Find $\sqrt{100}$.

$$\sqrt{100}$$

→ $\text{Side} = \sqrt{\text{Area}} = \sqrt{100} = 10 \text{ ft.}$

Answer: $\sqrt{100} = 10$

29. Evaluate: $\sqrt{169}$.

$$\sqrt{169}$$

→ Find the number that multiplied by itself equals 169.

→ $\sqrt{169} = 13$.

Answer: $\sqrt{169} = 13$



30. The area of a square is 9 sq ft. What is the side length? Find $\sqrt{9}$.

$$\sqrt{9}$$

$$\rightarrow \text{Side} = \sqrt{\text{Area}} = \sqrt{9} = 3 \text{ ft.}$$

Answer: $\sqrt{9} = 3$

